# A centre and a perpendicular

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Let $ABC$ be a triangle and $D$ a point of $[AB]$. The circumcircle of $BCD$ meets $(AC)$ again at $M$ and the circumcircle of $ACD$ meets $(BC)$ again at $N$. Let $O$ be the circumcentre of $CMN$. Show that $(OD)$ is perpendicular to $(AB)$.